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*Proposers may select any numbering mechanism for the proposal. The entire proposal however, must be paginated. Complete both columns only if the proposal is numbered consecutively.

COVER SHEET FOR PROPOSAL TO THE NATIONAL SCIENCE FOUNDATION

PROGRAM ANNOUNCEMENT/SOLICITATION NO./DUE DATE NSF 26-506 09/01/2026		☐ Special Exception to Deadline Date Policy		FOR NSF USE ONLY NSF PROPOSAL NUMBER	
FOR CONSIDERATION BY NSF ORGANIZATION UNIT(S) (Indicate the most specific unit known, i.e. program, division, etc.) TI - POSE					
DATE RECEIVED	NUMBER OF COPIES 1	DIVISION ASSIGNED 15030000 TI	FUND CODE 211Y00	UEI(Unique Entity Identifier) ZD93BGM5GXA5	FILE LOCATION
EMPLOYER IDENTIFICATION NUMBER (EIN) OR TAXPAYER IDENTIFICATION NUMBER (TIN) 874332677		SHOW PREVIOUS AWARD NO. IF THIS IS ☐ A RENEWAL ☐ AN ACCOMPLISHMENT-BASED RENEWAL		IS THIS PROPOSAL BEING SUBMITTED TO ANOTHER FEDERAL AGENCY? YES ☐ NO ☒ IF YES, LIST ACRONYM(S)	
NAME OF ORGANIZATION TO WHICH AWARD SHOULD BE MADE FREEMOCAP FOUNDATION, INC			ADDRESS OF AWARDEE ORGANIZATION, INCLUDING 9 DIGIT ZIP CODE 28 WENHAM ST APT 3 BOSTON,MA 02130-4119 US		
AWARDEE ORGANIZATION CODE (IF KNOWN) 7203601738					
NAME OF PRIMARY PLACE OF PERF FREEMOCAP FOUNDATION, INC			ADDRESS OF PRIMARY PLACE OF PERF, INCLUDING 9 DIGIT ZIP CODE 28 WENHAM ST APT 3 BOSTON,MA 02130-4119 US		
IS AWARDEE ORGANIZATION (Check All That Apply)		☐ SMALL BUSINESS ☐ FOR-PROFIT ORGANIZATION		☐ MINORITY BUSINESS ☐ WOMAN-OWNED BUSINESS	
					☐ IF THIS IS A PRELIMINARY PROPOSAL THEN CHECK HERE
TITLE OF PROPOSED PROJECT PESOSE: Track 1: Scoping a Sustainable and Secure Open-Source Ecosystem for the Free Motion Capture (FreeMoCap) Project					SHOW LETTER OF INTENT ID IF APPLICABLE
REQUESTED AMOUNT \$ 298,770	PROPOSED DURATION (1-60 MONTHS) 12 months	REQUESTED STARTING DATE 04/01/2027	SHOW RELATED PRELIMINARY PROPOSAL NO. IF APPLICABLE		
THIS PROPOSAL INCLUDES ANY OF THE ITEMS LISTED BELOW					
☒ TYPE OF PROPOSAL Research ☒ COLLABORATIVE STATUS Non-Collaborative ☐ BEGINNING INVESTIGATOR ☐ DISCLOSURE OF LOBBYING ACTIVITIES ☐ PROPRIETARY & PRIVILEGED INFORMATION ☐ HISTORIC PLACES ☐ LIVE VERTEBRATE ANIMALS IACUC App. Date _____ PHS Animal Welfare Assurance Number _____			☐ HUMAN SUBJECTS Human Subjects Assurance Number _____ Exemption Subsection _____ or IRB App. Date _____ ☐ FUNDING OF INT'L BRANCH CAMPUS OF U.S. IHE ☐ FUNDING OF FOREIGN ORGANIZATION OR FOREIGN INDIVIDUAL ☐ INTERNATIONAL ACTIVITIES: COUNTRY/COUNTRIES INVOLVED _____ ☐ POTENTIAL LIFE SCIENCES DUAL USE RESEARCH OF CONCERN ☐ OFF-CAMPUS OR OFF-SITE RESEARCH ☐ POTENTIAL IMPACTS ON INDIAN TRIBES		
PI/PD DEPARTMENT		PI/PD POSTAL ADDRESS 28 Wenham St #3 Boston,MA 021304119 US			
PI/PD FAX NUMBER					
NAMES(TYPED)	High Degree	Yr of Degree	Telephone Number	EmailAddress	
PI/PD NAME Jonathan S Matthis	PhD	2014	503-449-6999	jon@freemocap.org	
CO-PI/PD Aaron T Cherian	PhD	2026	845-269-9563	aaron@freemocap.org	
CO-PI/PD					
CO-PI/PD					
CO-PI/PD					

CERTIFICATION PAGE

Certification for Authorized Organizational Representative (or Equivalent)

By electronically signing and submitting this proposal, the Authorized Organizational Representative (AOR) is: (1) certifying that statements made herein are true and complete to the best of the individual's knowledge; and (2) agreeing to accept the obligation to comply with NSF award terms and conditions if an award is made as a result of this proposal. Further, the proposer is hereby providing certifications regarding conflict of interest, flood hazard insurance, responsible and ethical conduct of research, organizational support, and safe and inclusive working environments for off-campus or off-site research, as set forth in the NSF Proposal & Award Policies & Procedures Guide (PAPPG). Willful provision of false information in this application and its supporting documents or in reports required under an ensuing award is a criminal offense (U.S. Code, Title 18, Section § 1001).

Certification Regarding Conflict of Interest

The AOR is required to complete certifications stating that the organization has implemented and is enforcing a written policy on conflicts of interest (COI), consistent with the provisions of PAPPG Chapter IX.A.; that, to the best of the individual's knowledge, all financial disclosures required by the conflict of interest policy were made; and that conflicts of interest, if any, were, or prior to the organization's expenditure of any funds under the award, will be, satisfactorily managed, reduced or eliminated in accordance with the organization's conflict of interest policy. Conflicts that cannot be satisfactorily managed, reduced or eliminated and research that proceeds without the imposition of conditions or restrictions when a conflict of interest exists, must be disclosed to NSF via use of the Notifications and Requests Module in Research.gov.

Certification Regarding Flood Hazard Insurance

Two sections of the National Flood Insurance Act of 1968 (42 U.S.C § 4012a and § 4106) bar Federal agencies from giving financial assistance for acquisition or construction purposes in any area identified by the Federal Emergency Management Agency (FEMA) as having special flood hazards unless the: (1) community in which that area is located participates in the national flood insurance program; and (2) building (and any related equipment) is covered by adequate flood insurance. By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) located in FEMA-designated special flood hazard areas is certifying that adequate flood insurance has been or will be obtained in the following situations: (1) for NSF awards for the construction of a building or facility, regardless of the dollar amount of the award; and (2) for other NSF awards when more than \$25,000 has been budgeted in the proposal for repair, alteration or improvement (construction) of a building or facility.

Certification Regarding Responsible and Ethical Conduct of Research (RECR)

(Applies to Institutions of Higher Education only and is not applicable to proposals for conferences, symposia, and workshops)

By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that the institution has a plan in place to provide appropriate training and oversight in the responsible and ethical conduct of research to undergraduate students, graduate students, postdoctoral researchers, faculty, and other senior/key personnel who will be supported by NSF to conduct research and that such training addresses mentor training and mentorship, training to raise awareness of potential research security threats, and Federal export control, disclosure, and reporting requirements.

The AOR shall require that the language of this certification be included in any award documents for all subawards at all tiers.

Certification Regarding Organizational Support

By electronically signing the Certification Pages, the Authorized Organizational Representative (or equivalent) is certifying that there is organizational support for the proposal as required by Section 526 of the America COMPETES Reauthorization Act of 2010. This support extends to the portion of the proposal developed to satisfy the Broader Impacts Review Criterion as well as the Intellectual Merit Review Criterion, and any additional review criteria specified in the solicitation. Organizational support will be made available, as described in the proposal, in order to address the broader impacts and intellectual merit activities to be undertaken.

Certification Regarding Dual Use Research of Concern

By electronically signing the certification pages, the Authorized Organizational Representative is certifying that the organization will be or is in compliance with all aspects of the United States Government Policy for Institutional Oversight of Life Sciences Dual Use Research of Concern.

Certification Requirement Specified in the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, Section 223(a)(1) (42 USC 6605(a)(1))

By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that each individual employed by the organization and identified on the proposal as senior/key personnel has been made aware of the certification requirements identified in the William M. (Mac) Thornberry National Defense Authorization Act for Fiscal Year 2021, Section 223(a)(1) (42 USC 6605(a)(1)).

Certification Regarding Safe and Harassment-free Working Environments for Off-Campus or Off-Site Research

(This certification applies only to proposals in which data/information/samples are being collected off-campus or off-site, such as fieldwork and research activities on vessels and aircraft.) By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in accordance with the NSF Proposal & Award Policies and Procedures Guide, Chapter II.E.9, the organization has a plan in place for this proposal regarding safe and harassment-free working environments.

Certification Regarding Malign Foreign Talent Recruitment Programs

By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. 19232), all individuals identified as senior/key personnel have been made aware of and have complied with their responsibility under that section to certify that the individual is not a party to a malign foreign talent recruitment program. Additionally, individuals serving as a Principal Investigator (PI) or co-PI on an active NSF award made on or after May 20, 2024, are aware they must certify annually via Research.gov to their participation or non-participation in an MFTRP. Additional information regarding the annual certification process can be found at nsf.gov/research-security.

Certification Regarding Research Security Training for Federal Research Award Personnel

By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), the AOR is certifying that all individuals identified as senior/key personnel have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certification Regarding IHEs Hosting or Supporting Confucius Institutes

By electronically signing the Certification Pages, the Authorized Organizational Representative is certifying that, in compliance with Section 10339A of the CHIPS and Science Act of 2022 (42 U.S.C. § 19039), the AOR is certifying that, absent a waiver granted by the NSF Director, the IHE does not maintain a contract or agreement between the institution and a Confucius Institute.

AUTHORIZED ORGANIZATIONAL REPRESENTATIVE		SIGNATURE	DATE
NAME			
TELEPHONE NUMBER	EMAIL ADDRESS	FAX NUMBER	

IDENTIFYING INFORMATION:

NAME: Matthis, Jonathan Samir

ORCID iD: <https://orcid.org/0000-0003-3683-646X>

POSITION TITLE: President

PRIMARY ORGANIZATION AND LOCATION: FreeMoCap Foundation, Inc. , Boston, Massachusetts, United States**Professional Preparation:**

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
University of Texas at Austin, Austin, Texas, United States	Postdoctoral Fellow	09/2014 - 06/2019	Perceptual Systems
Rensselaer Polytechnic Institute, Troy, New York, United States	PHD	05/2011 - 05/2014	Cognitive Science
Rensselaer Polytechnic Institute, Troy, New York, United States	MS	09/2008 - 05/2011	Cognitive Science
University of Maryland Baltimore County, Baltimore, MD, United States	BA	09/2005 - 05/2007	Philosophy

Appointments and Positions

2021 - present President, FreeMoCap Foundation, Inc. , Boston, Massachusetts, United States

2021 - present President/CEO, FreeMoCap Foundation, Inc., Boston, Massachusetts, United States

2019 - 2026 Assistant Professor, Northeastern University, Boston, Massachusetts, United States

Products**Products Most Closely Related to the Proposed Project**

1. Matthis JS, Yates JL, Hayhoe MM. Gaze and the Control of Foot Placement When Walking in Natural Terrain. Curr Biol. 2018 Apr 23;28(8):1224-1233.e5. PubMed Central PMCID: [PMC5937949](https://pubmed.ncbi.nlm.nih.gov/PMC5937949/).
2. Matthis JS, Barton SL, Fajen BR. The critical phase for visual control of human walking over complex terrain. Proc Natl Acad Sci U S A. 2017 Aug 8;114(32):E6720-E6729. PubMed Central PMCID: [PMC5558990](https://pubmed.ncbi.nlm.nih.gov/PMC5558990/).
3. Jonathan Samir Matthis, Philip Queen, Aaron Cherian, Trent Wirth, Endurance Idehen. FreeMoCap: A free, open source markerless motion capture system. 2026 August. Available from: <https://zenodo.org/doi/10.5281/zenodo.22131400> DOI: 10.5281/ZENODO.22131400
4. jonmatthis, Endurance Idehen, philipqueen, Mac Prible, Aaron Cherian. SkellyCam: The camera backend for the FreeMoCap software. 2026 April. Available from:

<https://zenodo.org/doi/10.5281/zenodo.19921107> DOI: 10.5281/ZENODO.19921107

5. Matthis JS, Muller KS, Bonnen KL, Hayhoe MM. Retinal optic flow during natural locomotion. PLoS Comput Biol. 2022 Feb;18(2):e1009575. PubMed Central PMCID: [PMC8896712](https://pubmed.ncbi.nlm.nih.gov/PMC8896712/).

Other Significant Products, Whether or Not Related to the Proposed Project

1. Matthis JS, Barton SL, Fajen BR. The biomechanics of walking shape the use of visual information during locomotion over complex terrain. J Vis. 2015 Mar 18;15(3) PubMed Central PMCID: [PMC4365914](https://pubmed.ncbi.nlm.nih.gov/PMC4365914/).
2. Matthis JS, Fajen BR. Visual control of foot placement when walking over complex terrain. J Exp Psychol Hum Percept Perform. 2014 Feb;40(1):106-15. PubMed PMID: [23750964](https://pubmed.ncbi.nlm.nih.gov/23750964/); NIHMSID: NIHMS1050949.
3. philipqueen, jonmatthis, Aaron Cherian. SkellyClicker: For labeling and traing multi-camera video datasets. 2026 April. Available from: <https://zenodo.org/doi/10.5281/zenodo.19389889> DOI: 10.5281/ZENODO.19389889
4. philipqueen, jonmatthis. SkellyTracker: The image analysis backend for the FreeMoCap Project. 2023 December. Available from: <https://zenodo.org/doi/10.5281/zenodo.10395641> DOI: 10.5281/ZENODO.10395641
5. Aaron Cherian, jonmatthis. freemocap/skellyforge: 2023.12.1005. 2023 December. Available from: <https://zenodo.org/doi/10.5281/zenodo.10395649> DOI: 10.5281/ZENODO.10395649

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Matthis, Jonathan Samir in SciENCv on 2026-08-28 15:03:59

IDENTIFYING INFORMATION:

NAME: Cherian, Aaron

ORCID iD: <https://orcid.org/0000-0003-4944-2814>

POSITION TITLE: Chief Scientific Officer

PRIMARY ORGANIZATION AND LOCATION: FreeMoCap Foundation, Boston, Massachusetts, United States**Professional Preparation:**

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
Northeastern University, Boston, Massachusetts, United States	PHD	09/2021 - 05/2026	Biomedical Engineering
Brown University, Providence, Rhode Island, United States	MS	09/2018 - 05/2020	Biomedical Engineering
University of Maryland College Park, College Park, Maryland, United States	BS	09/2014 - 05/2018	Bioengineering

Appointments and Positions

2026 - present Chief Scientific Officer, FreeMoCap Foundation, Boston, Massachusetts, United States

Products**Products Most Closely Related to the Proposed Project**

1. Cherian AT. Open-Source Development and Validation of a Low-Cost Markerless System for Quantitative Motion Analysis. Boston, Massachusetts: Northeastern University; 2026 May. Available from: <https://hdl.handle.net/2047/D20859645>
2. Jonathan Samir Matthis, Philip Queen, Aaron Cherian, Trent Wirth, Endurance Idehen. FreeMoCap: A free, open source markerless motion capture system. 2026 August. Available from: <https://zenodo.org/doi/10.5281/zenodo.22131400> DOI: 10.5281/ZENODO.22131400
3. Aaron Cherian, jonmatthis. SkellyForge: The 3D data processing backend for the FreeMoCap Project. 2023 December. Available from: <https://zenodo.org/doi/10.5281/zenodo.10395649> DOI: 10.5281/ZENODO.10395649
4. philipqueen, jonmatthis, Aaron Cherian. SkellyClicker: For labeling and traing multi-camera video datasets. 2026 April. Available from: <https://zenodo.org/doi/10.5281/zenodo.19389889> DOI: 10.5281/ZENODO.19389889
5. jonmatthis, Endurance Idehen, philipqueen, Mac Prible, Aaron Cherian. SkellyCam: The camera backend for the FreeMoCap software. 2026 April. Available from: <https://zenodo.org/doi/10.5281/zenodo.19921107> DOI: 10.5281/ZENODO.19921107

Other Significant Products, Whether or Not Related to the Proposed Project

1. Nie J, Eom K, AlGhosain HM, Neifert A, Cherian A, Gerbaka GM, Ma KY, Liu T, Lee J.

Intravitreally Injected Plasmonic Nanorods Activate Bipolar Cells with Patterned Near-Infrared Laser Projection. ACS Nano. 2025 Apr 1;19(12):11823-11840. PubMed Central PMCID: [PMC12105531](https://pubmed.ncbi.nlm.nih.gov/PMC12105531/).

2. philipqueen, jonmatthis. SkellyTracker: The image analysis backend for the FreeMoCap Project. 2023 December. Available from: <https://zenodo.org/doi/10.5281/zenodo.10395641> DOI: 10.5281/ZENODO.10395641
3. Aaron Thomas Cherian. Development and Optimization of an Intrinsic Optical Imaging System for Hemodynamic Correction of Fluorescent Signals. 2020 May. Available from: <https://repository.library.brown.edu/studio/item/bdr:1129219/> DOI: 10.26300/XHJ5-KV66

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Cherian, Aaron in SciENCv on 2026-08-31 11:34:41

Table 1

1	Your Name:	Your Organizational Affiliation(s), last 12 mo	Last Active Date
	Matthis, Jonathan Samir	FreeMoCap Foundation, Inc., Boston, MA	
		Northeastern University, Boston, MA	06/29/26

Table 2

2	Name:	Type of Relationship	Optional (email, Department)	Last Active Date
R	Matthis, Paul	Family (brother)		
R	Cherian, Aaron	Business (co-PI, FreeMoCap Foundation)		

Table 3

3	Advisor/Advisee Name:	Organizational Affiliation	Optional (email, Department)
G	Fajen, Brett R.	Rensselaer Polytechnic Institute, Troy, NY	
T	Cherian, Aaron	Northeastern University, Boston, MA	

Table 4

4	Name:	Organizational Affiliation	Optional (email, Department)	Last Active Date
A	Hayhoe, Mary M.	University of Texas Austin		
A	Muller, Karl S.			
A	Bonnen, Kathryn	Indiana University Bloomington		
A	Panfili, Daniel P.			
A	Shields, Stephanie M.			
A	Cormack, Lawrence K.	University of Texas Austin		
A	Huk, Alexander C.	UCLA		
A	Powell, Nathaniel V.			
A	Oh, Youjin			
A	Cherian, Aaron	Northeastern University / FreeMoCap Foundation, Boston, MA		
A	Wirth, Trenton D.	U Cincinnati		
A	Queen, Philip	CU Denver Anschutz		
A	Idehen, Endurance			
C	Bayus, David	San Jose State University, CADRE Laboratory for New		

		Media, San Jose, CA		
C	Nilsson, Jonathan	GIBBET GAMES AB, Skovde, Sweden		
C	Tukes, Raven	The Possible Zone, Boston, MA		
C	Kerr, Sarah	Imatest LLC, Boulder, CO		
C	Casals, Andres			
C	Scott, Dominic			
C	Scholl, Ben	CU Denver Anschutz		

Table 5

5	Name:	Organizational Affiliation	Journal/Collection	Last Active Date

Table 1

1	Your Name:	Your Organizational Affiliation(s), last 12 mo	Last Active Date
	Cherian, Aaron	FreeMoCap Foundation, Inc., Boston, MA	
		Northeastern University, Boston, MA	04/29/26

Table 2

2	Name:	Type of Relationship	Optional (email, Department)	Last Active Date
R	Matthis, Jonathan Samir	Business (PI, FreeMoCap Foundation); PhD advisor		

Table 3

3	Advisor/Advisee Name:	Organizational Affiliation	Optional (email, Department)
G	Matthis, Jonathan Samir	Northeastern University, Boston, MA	

Table 4

4	Name:	Organizational Affiliation	Optional (email, Department)	Last Active Date
A	Nie, Jiarui			
A	Eom, Kyungsik			
A	AlGhosain, Hafithe M.			
A	Neifert, Alexander			
A	Gerbaka, Gaia Marie			
A	Ma, Kristine Y.			
A	Liu, Tao			
A	Lee, Jonghwan	Brown University, Providence, RI		
A	Matthis, Jonathan Samir	FreeMoCap Foundation, Inc., Boston, MA		
A	Queen, Philip			
A	Wirth, Trenton D.			
A	Idehen, Endurance			
C	Bayus, David	San Jose State University, CADRE Laboratory for New Media, San Jose, CA		
C	Nilsson, Jonathan	GIBBET GAMES AB, Skovde, Sweden		
C	Tukes, Raven	The Possible Zone, Boston, MA		

C	Kerr, Sarah	Imatest LLC, Boulder, CO		
C	Casals, Andres			
C	Scott, Dominic			
C	Scholl, Ben			

Table 5

5	Name:	Organizational Affiliation	Journal/Collection	Last Active Date

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Matthis, Jonathan Samir

PERSISTENT IDENTIFIER (PID) OF THE SENIOR/KEY PERSON: <https://orcid.org/0000-0003-3683-646X>

*POSITION TITLE: President

*ORGANIZATION AND LOCATION: FreeMoCap Foundation, Inc., Boston, Massachusetts, United States

Proposals/Active Projects

*Proposal/Active Project Title: PESOSE: Track 1: Scoping a Sustainable and Secure Open-Source Ecosystem for the Free Motion Capture (FreeMoCap) Project

*Status of Support: Pending

Proposal/Award Number:

*Source of Support: National Science Foundation

*Primary Place of Performance: FreeMoCap Foundation, Boston, MA, USA

*Proposal/Active Project Start Date: (MM/YYYY): 04/2027

*Proposal/Active Project End Date: (MM/YYYY): 04/2028

*Total Anticipated Proposal/Project Amount: \$298,770

* Person Months per budget period Devoted to the Proposal/Active Project:

Year	Person Months
2027	8

***Overall Objectives:**

This Track 1 project will scope the organizational, technical, and community structures required to transition FreeMoCap, a free and open-source markerless motion-capture system used by more than 15,000 people across 153 countries, from a project maintained by a small core team into a durable open-source ecosystem with distributed contributors and shared stewardship.

Planned activities are: (1) ecosystem discovery mapping the existing user and contributor base through the mandatory I-Corps for PESOSE training; (2) documentation and educational pathways supporting progression from independent use to informed contribution; (3) governance and licensing models that distribute technical decision-making beyond the founding team; (4) a community plug-in framework with associated review, quality, and security controls; (5) a reproducible validation and benchmarking methodology designed for independent reproduction by outside laboratories; and (6) models for long-term financial and community sustainability.

The objective is an ecosystem in which contributors outside the founding team can independently extend the

software, participate in technical decision-making, and share responsibility for its validation and maintenance. Track 1 supports planning and scoping for this transition, not product development.

***Statement of Potential Overlap:**

None

Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Matthis, Jonathan in SciENcv on 2026-08-31 17:51:51

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Cherian, Aaron T

PERSISTENT IDENTIFIER (PID) OF THE SENIOR/KEY PERSON: <https://orcid.org/0000-0003-4944-2814>

*POSITION TITLE: Chief Scientific Officer

*ORGANIZATION AND LOCATION: FreeMoCap Foundation, Boston, Massachusetts, United States

Proposals/Active Projects

***Proposal/Active Project Title:** PESOSE: Track 1: Scoping a Sustainable and Secure Open-Source Ecosystem for the Free Motion Capture (FreeMoCap) Project

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** FreeMoCap Foundation, Boston, MA., USA

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2027

***Proposal/Active Project End Date: (MM/YYYY):** 04/2028

***Total Anticipated Proposal/Project Amount:** \$298,770

* **Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	9

***Overall Objectives:**

This Track 1 project will scope the organizational, technical, and community structures required to transition FreeMoCap, a free and open-source markerless motion-capture system used by more than 15,000 people across 153 countries, from a project maintained by a small core team into a durable open-source ecosystem with distributed contributors and shared stewardship.

Planned activities are: (1) ecosystem discovery mapping the existing user and contributor base through the mandatory I-Corps for PESOSE training; (2) documentation and educational pathways supporting progression from independent use to informed contribution; (3) governance and licensing models that distribute technical decision-making beyond the founding team; (4) a community plug-in framework with associated review, quality, and security controls; (5) a reproducible validation and benchmarking methodology designed for independent reproduction by outside laboratories; and (6) models for long-term financial and community sustainability.

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software, participate in technical decision-making, and share responsibility for its validation and maintenance. Track 1 supports planning and scoping for this transition, not product development.

***Statement of Potential Overlap:**

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Certification:

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

In accordance with Section 10632 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19232), each individual identified as a senior/key person must certify that they are not a party to a malign foreign talent recruitment program.

Research Security Training Requirement for Federal Award Personnel: In accordance with Section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), each individual identified as a senior/key person must certify that they have completed the requisite research security training that meets the requirements specified in Item 2 of Important Notice No. 149 within 12 months prior to proposal submission.

Certified by Cherian, Aaron in SciENCv on 2026-08-31 18:06:03

Overview

FreeMoCap is a free and open-source markerless motion-capture system that turns synchronized video from consumer-grade webcams into research-grade full-body kinematic data using AI-based pose estimation, bringing quantitative movement measurement within reach of laboratories, clinics, and classrooms for which conventional systems are not an option. Since 2021, FreeMoCap has grown to more than 15,000 users across 153 countries, including researchers, artists, students, educators, and technologists. However, this growth has not been matched by a corresponding distribution of the knowledge and responsibility needed to sustain it. The understanding required to modify the software, the work of user support, and authority over technical decisions all remain concentrated in a small core team. Without organizational change, the weight of supporting a large user base will constrain both the project's development and its long-term viability.

This Track 1 project will scope the structures needed to transition FreeMoCap into a durable open-source ecosystem with distributed contributors and shared stewardship. Activities include user base mapping through I-Corps for PESOSE, documentation and educational pathways for independent users and contributors, governance models for distributing technical decision-making, a community plug-in framework with review and security controls, reproducible validation and benchmarking, and models for long-term sustainability. The long-term goal is an ecosystem in which contributors outside the founding team can independently extend the software, participate in technical decision-making, and share responsibility for its validation and maintenance.

Intellectual Merit

The intellectual merit of this project lies in testing whether organizational structures and documented contributor pathways can reduce duplicated effort in scientific software by making it easier for researchers to contribute specialized capabilities to shared infrastructure rather than rebuild complete pipelines from scratch. The project also addresses a challenge particularly relevant to scientific software, in which software modifications can silently alter measured outputs without producing any obvious failure. The stage-wise integrity and benchmarking methodology scoped here will be designed for reproduction by outside laboratories on their own hardware. Together, these activities will establish how knowledge decentralization, community plug-in governance, and reproducible measurement-integrity practices can support the transition from isolated scientific software development to shared research infrastructure.

Broader Impacts

Quantitative movement measurement supports work across many disciplines, yet the instrumentation and technical expertise required to obtain it remain out of reach for many. FreeMoCap lowers the hardware barrier through free and open-source software that runs on consumer-grade cameras, while the ecosystem scoped under this project will address the knowledge barrier through educational and onboarding pathways from independent use to informed contribution. A shared, extensible ecosystem also allows methods and expertise developed in one community to become useful in others rather than remain isolated in field-specific software or laboratory pipelines. Educational activities piloted with partner organizations, including The Possible Zone, will test materials for users with differing technical backgrounds. By broadening access, technical understanding, and cross-domain participation, the project can increase the return on public investment in scientific software while expanding who can use, understand, and extend quantitative movement-measurement tools.

Keywords: open-source ecosystem; motion capture; biomechanics; research software sustainability

Vision and Need

FreeMoCap [1] is a free and open-source markerless motion capture software system that produces research-grade full body kinematic data from generic cameras (Figure 1). Built and maintained by the FreeMoCap Foundation since 2021, it has grown organically into a vibrant, successful project with over 10,000 GitHub stars, telemetry indicating at least 15,000 users across 153 countries, and a global community forum of over 4,000 students, researchers, educators, and artists.

The project’s broad adoption and continued growth reveal a real and unmet need. Quantitative movement measurement supports work across biomechanics, neuroscience, rehabilitation, and animation, yet conventional systems remain costly and are often tied to specialized hardware or fixed workflows. Researchers whose needs fall outside those workflows therefore continue to rebuild substantial portions of motion-capture infrastructure independently. By running on consumer-grade cameras and following established patterns for the development and management of free-and-open-source-software (FOSS), FreeMoCap provides a validated, research-grade measurement tool appropriate for high-level research laboratories as well as classrooms, garage labs, and basement mocap studios [2].

FreeMoCap now sits at a critical inflection point in its growth curve. As is common for mid-scale FOSS projects, our user count has grown far faster than its maintainer base, a community pattern that Nadia Eghbal’s study of open source software, *Working In Public*, refers to as a “Stadium” [3] - many users relying on the output of a proportionally tiny team of developers. If growth in the user base continues to outpace its maintainers, the growing weight of support runs the risk of overwhelming the core team. Sustaining the project through this period requires building the organizational and community infrastructure that supports a transition toward what Eghbal calls a “Federation,” a pattern representing a healthy balance of user and contributor growth, allowing technical knowledge and responsibility to be distributed across the community.

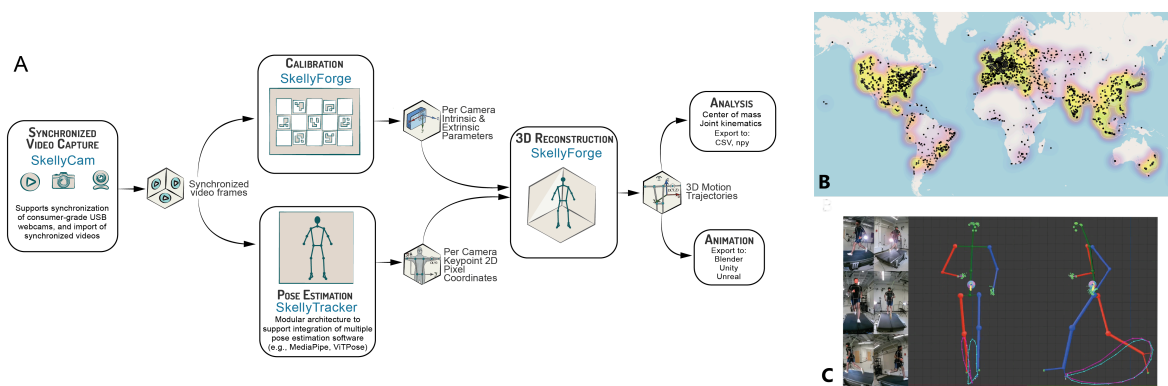


Figure 1: (A) The FreeMoCap workflow and its component repositories (blue). (B) Global FreeMoCap usage. (C) Reconstructed 3D trajectories alongside the source camera views.

Despite this growth, our understanding of the broader FreeMoCap community remains limited. Telemetry and interactions with users who participate in public forums do not reliably tell us who our users are, what they need, or where existing technical and community infrastructure create barriers.

This proposal requests Track 1 support to scope the foundation for future growth: understanding our **community landscape** and its needs; developing **documentation and upskilling pathways** that convert user interest into contributor capacity; governance practices that create **transparency and productive communication** between the core maintainers and the wider

community; a **validation and benchmarking methodology** that other laboratories can replicate; and a **sustainability model** for the maintenance that the software will require.

Current State of the Software and Community

The FreeMoCap codebase is hosted in a collection of public GitHub repositories under the freemocap organization ([github-dot-com-slash-freemocap](https://github.com/freemocap)), built and tested through automated CI/CD pipelines, and distributed as a cross-platform OS-certified standalone desktop application via installers hosted through our website (freemocap-dot-org-slash-download).

The project is organized as a polyrepo - the core functionality of the application is split across domain-scoped constituent sub-repositories — *skellycam* for camera synchronization and recording, *skellytracker* as a modular interface for pose estimation, *skellyforge* for calibration, 3D reconstruction, and kinematic analysis, and *freemocap_blender_addon* for export to Blender (and other animation software) (Figure 1). Each sub-repository represents an independently functional codebase and follows the same internal layout, so that documentation, tests, and entry points sit in the same place in every repository. This self-similar structure makes the full codebase substantially easier to teach, to learn, and to navigate with AI coding assistants, and a contributor with expertise in one domain can work inside a single component without holding the whole system in mind.

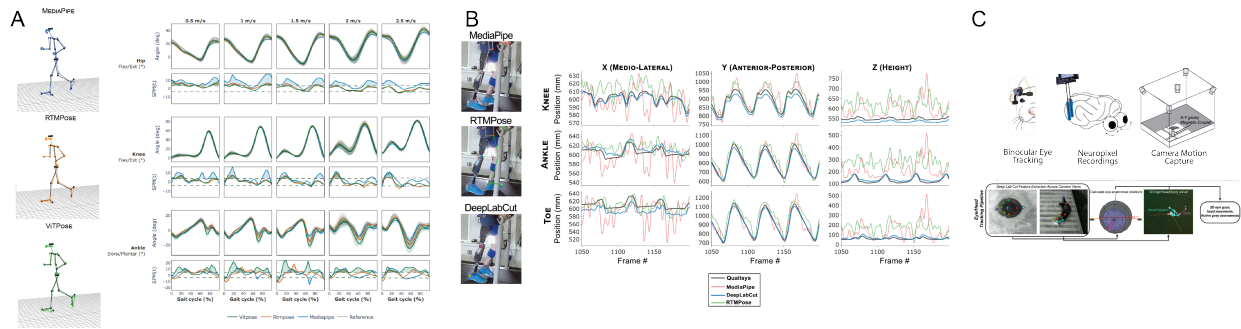


Figure 2: FreeMoCap applications from standard human biomechanics to specialized movement-analysis use cases. A. Lower-extremity joint-angle comparison and statistical parametric mapping against Qualisys marker-based motion capture across multiple pose-estimation backends. B. Custom DeepLabCut integration for tracking a lower-limb prosthesis user. C. Extension to ferret locomotion and neural-behavioral research in collaboration with the Scholl Lab, CU Denver Anschutz

FreeMoCap is used across several distinct communities, including athletes, clinicians, educators and students, technical artists, and general technologists. Published use of FreeMoCap spans reaching and grasping biomechanics [4], gait analysis [5], computational linguistics [6–8], animation [9], and robotic exoskeleton design and verification [10]. Independent validation studies have reported agreement with marker-based motion capture across several movement tasks [11, 12], while our own work [2] has characterized gait and balance performance, demonstrated integration of multiple pose-estimation backends, and extended the workflow to non-human locomotion (Figure 2).

Despite growing external contributions, much of the architectural and operational knowledge required for core development remains concentrated within a core development team, and pathways for contributor onboarding remain immature. Track 1 scoping will ground the development of these mechanisms in evidence about the community they are intended to serve.

Team Qualifications

The FreeMoCap Foundation team combines two decades of the PI’s research experience in camera-based measurement of human perceptuo-motor behavior with nearly a decade of the co-PI’s, whose

doctoral dissertation is the development and validation of the FreeMoCap system itself [2]. The team has built, validated, distributed, and supported FreeMoCap and its community for more than five years, carrying it from a single-laboratory tool to an instrument used in 153 countries, and has managed the technical and organizational challenges that growth produced.

Ecosystem Discovery

Need for Innovation

A variety of motion capture tools currently exist, but each is built around particular assumptions about users, domains, hardware, or deployment. Commercial systems include biomechanics-focused platforms such as Theia3D and animation tools such as Move.ai and Rokoko, while open-source alternatives such as OpenCap [13], Pose2Sim [14], and PosePipe [15] are largely centered on applications of human biomechanics. These systems also impose different operational constraints: Theia3D relies on expensive hardware, while OpenCap’s standard workflow requires supported iOS devices, account log-in screens, and remote cloud processing.

Researchers with specialized requirements still build substantial portions of this pipeline independently, as evidenced by the continued development of domain-specific workflows in individual laboratories [16–19], often reimplementing the stages shown in Figure 1. Across projects, this duplicates technical effort and reduces the return on public research investment in software that is difficult to maintain, reproduce, or transfer beyond the environment in which it was developed.

Our Track 1 scoping will test the premise that the **repeated rebuilding of motion-capture pipelines is an ecosystem failure**, as researchers looking for extensible tools often find building from scratch easier than extending shared infrastructure. What is needed is not only accessible motion-capture software, but an ecosystem in which specialized adaptations developed by one community can be contributed back, maintained, and reused by others — infrastructure that depends on defined stewardship and governance as much as on code. Track 1 will determine what documentation, contribution pathways, technical interfaces, and governance structures are needed to reduce the barriers to extending and contributing to shared infrastructure.

Necessity of an open-source ecosystem approach

FreeMoCap’s existing community provides early evidence that an open-source ecosystem can convert individual expertise into shared infrastructure. For example, after adopting FreeMoCap for digital-media instruction and animation, collaborators at San José State University developed HIHO Mocap, an open-source Blender add-on that extends FreeMoCap output for classroom and animation workflows rather than rebuilding the underlying motion-capture pipeline. Similar transitions have occurred in UI design and GPU-accelerated pose-estimation architecture, bringing domain expertise from outside the core development team into the ecosystem for the benefit of other users. However, these transitions occurred informally and have depended heavily on direct interaction with the core team. Creating a repeatable path into the ecosystem first requires understanding **who** is trying to use, teach, extend, or contribute to FreeMoCap, **what** technical knowledge each group needs, and **where** the current software and documentation fail to support them independently.

Stakeholder discovery

Our current understanding of the FreeMoCap community is drawn primarily from users who contact us directly or participate in existing community channels. Some applications have emerged outside the domains we originally anticipated. For example, computational-linguistics researchers have adopted FreeMoCap in published work [6–8] without any prior interaction with the core team. This motivates a broader discovery effort aimed not only at *known users*, but also at

understanding unexpected communities, what drew them to the platform, and what would reduce barriers to sustained and broader adoption.

Our proposed work will combine analysis of existing community activity, community-wide surveys, and the I-Corps for PESOSE stakeholder-discovery process to characterize four user populations reflecting the communities already observed around FreeMoCap: (1) **researchers** using FreeMoCap as part of scientific projects, (2) **educators** building FreeMoCap-based teaching materials, (3) **technical artists** working in 3D animation and related fields, and (4) current and prospective **contributors**. Discovery will produce three assessments, each addressing one condition under which extending shared infrastructure becomes easier than rebuilding it.

1) **Cross-domain needs assessment**: how different communities use FreeMoCap, what brought them to it, and what their work requires that our standard pipelines do not yet provide.

2) **Knowledge assessment**: what users must understand to *operate* the software, what contributors must understand to *modify* it, which of that knowledge remains undocumented, and what forms of documentation or instruction would allow people to participate without direct access to the core team.

3) **Comparative ecosystem review**: how mature open-source projects of comparable or larger scale structure contribution, extension, and decision-making, and how those structures perform as projects grow. These findings will supply candidate models for evaluation in *Organization and Governance* and inform the security, community-building, and sustainability work that follows.

Organization and Governance

Governance Scoping

Healthy open source ecosystems employ methods to distribute decision-making and knowledge such that it does not depend on any one person. In the Python Enhancement Proposal (PEP) process, upcoming software changes are announced and discussed via public Request For Comments (RFCs) that invite community input but (critically) do not gate progress on strict consensus. NumPy and SciPy run comparable processes (NEP and SEP, respectively) for stakeholder-facing transparency. What these share is a way of keeping consequential decisions visible and open to comment without making every decision wait on consensus. Track 1 will determine how these patterns should be adapted for a project of FreeMoCap’s scale, shaped by the community landscape revealed through ecosystem discovery.

Distributed infrastructure and community plug-ins

FreeMoCap’s modular architecture provides a technical basis for community plug-ins. Because the component repositories are organized around fixed domains — cameras, calibration, image analysis, reconstruction — rather than around any particular model or library, a new pose-estimation method can be adopted without disturbing the rest of the pipeline. For example, the `skellytracker` architecture exposes an open interface through which AI-based pose-estimation models developed by different groups can interoperate with a shared scientific workflow without modification to the underlying pipeline (Figure 2 A). This interface matters because no single pose-estimation model is adequate across all subjects. The algorithms in common use are trained on standard human datasets, and the requirements of biomechanical measurement differ from those the models were built to satisfy [20].

A **community plug-in**, in this context, is code written and maintained by someone outside the core team that runs against a published interface without altering the shared pipeline. A researcher whose subjects fall outside existing models could implement a tracker against a documented interface rather than assembling an entire pipeline around it, and the result would be available to

everyone facing the same problem, while an AI-tracking method published by a computer-vision researcher can be translated into a reusable scientific workflow for users who would otherwise lack the technical expertise to implement it.

A community plug-in system is a classic protection against “scope creep” - the ecosystem can grow in capability without the core team absorbing the maintenance of every addition, and users can adopt new capabilities without waiting for the core maintainers to implement them. Extensions that prove useful additions through community use can be pulled into the core. Community plug-ins allow for a separation between the development of the core architecture (open only to high-level developers and trusted contributors) and the softer, lower-stakes context of plug-in development.

Track 1 will define the review, distribution, maintenance, and responsibility structures required for this Community Plug-In model, informed by projects already operating plug-in ecosystems and the needs and preferences of the members of our community who are likely to participate.

Licensing

FreeMoCap and its component repositories are licensed under the AGPLv3+. Because so much of the landscape of closed-source motion capture operates behind proprietary servers, the AGPLv3+ is the only standard license whose terms would stop a closed-source competitor from running FreeMoCap behind a server endpoint while charging users for access. Because the Foundation holds the rights to the codebase, it can also offer alternative terms to organizations whose intended use is incompatible with the AGPLv3+, keeping the software freely available to everyone else while generating resources for maintenance and stewardship.

Long-term Sustainability

The FreeMoCap Foundation was incorporated as an IRS-recognized 501(c)(3) public charity to provide an institutional home for the project. In parallel, the Foundation will evaluate a diversified model for the long-term sustainability of this research infrastructure, including alternative commercial licensing, standardized hardware kits geared towards students, with higher-end software targeting more resourced labs, training and support services, institutional partnerships, and continued grant funding. SkellyTechnologies, LLC, a separate for-profit entity associated with the project, provides an additional pathway for commercial services, SBIR/STTR proposals, B2B agreements, and other profit-seeking activities that may support the long-term sustainability of the ecosystem.

Risk Analysis and Security Plan

Video-based motion capture creates inherent privacy concerns because identifiable recordings must be captured and processed. These concerns can themselves become a barrier to adoption: recent work on OpenCap found that patients remained concerned about video sharing, storage, and access despite HIPAA-compliant cloud safeguards [21]. FreeMoCap avoids this cloud-data surface by processing video and derived data **locally**, without requiring a login or network connection, allowing sensitive recordings to remain under the user’s direct control.

Secure and trustworthy research infrastructure also requires addressing **software-supply-chain** risk. FreeMoCap depends on a substantial stack of computer-vision, machine-learning, and scientific-computing libraries, and the proposed community plug-in system would introduce additional third-party code into the processing environment. Track 1 will assess FreeMoCap against security standards maintained by the Open Source Security Foundation (OpenSSF), which provides established criteria and certification pathways for secure open-source development. This assessment will identify gaps and the requirements needed to reach an appropriate certification level, informing

both a project security roadmap and the review, approval, and release controls needed to guard against malicious or inadvertently unsafe community plug-ins.

Data Integrity

While the fidelity of FreeMoCap-produced data has been and continues to be validated, **preserving the integrity of that data (i.e., that modifications to the software do not silently alter the measurements it produces) is an ecosystem responsibility**. Track 1 will scope stage-specific integrity checks across the processing pipeline, with expected outputs tracked across software versions. Camera characteristics are also part of the measurement chain. Collaboration with Imatest LLC, whose work includes ISO/TC 42 standards activity on image-information content, will inform how imaging-system limitations can be characterized, reported, and made visible to users.

Because these are the same checks an outside laboratory would need in order to trust the software on its own data, Track 1 will also scope how we should package and communicate our validation and benchmarking methodology, so that other groups can reproduce our validation on their own hardware and contribute the results back as community validation.

Identity and Telemetry

Potential ecosystem features such as contributor recognition, authenticated accounts, and usage telemetry introduce additional privacy and security considerations. Track 1 will scope approaches that minimize collected data and preserve FreeMoCap's local-first principles, including evaluation of established authorization approaches such as OAuth 2.0 for (optional) account-based features. The project's industry mentor, Endurance Idehen, will advise this work, including authentication, authorization, telemetry, and related security controls. Together, these activities will produce a risk register, draft extension review and release controls, and a data-integrity and security roadmap.

Community Building

The FreeMoCap Foundation maintains an active Discord community forum of more than 4,000 members, structured so that users can ask and answer questions without relying exclusively on the core development team. The core team also hosts a weekly community call for project updates, user feedback, technical discussion, and interaction with potential contributors. These activities provide an established foundation for community engagement, but **rapid user growth in 2026 has increased the need for education and participation models that can scale beyond direct interaction with maintainers**.

Track 1 will develop and pilot accessible educational and participation pathways at three levels: **introductory materials** for general users; **intermediate and domain-specific resources** for scientific, technical, creative, and educational applications; and separate **developer documentation** covering software architecture, contribution workflows, testing, and maintenance.

These materials will be piloted through hands-on classes and workshops with partner organizations, including The Possible Zone and Artisans Asylum, and through classroom use with collaborators at San José State University, where FreeMoCap is already being used in digital-media instruction. These pilots will identify where learners encounter conceptual or technical barriers, which formats work best for different audiences, what reproducible hardware setups classroom deployment requires, and what knowledge can be transferred without direct core-team instruction.

Track 1 will also test low-barrier forms of community participation that do not require software development, including community-created visualizations, artwork, examples, challenges, and other ways for users to contribute domain expertise.

Together, these activities will inform a scalable education and onboarding model that supports progression from user, to informed participant, to sustained contributor. Longer term, the resulting model could inform a larger FreeMoCap community meeting or conference bringing together users, developers, educators, and researchers as part of a subsequent Track 2 effort.

Milestones and Evaluation Plan

Evaluation

Track 1 success will be evaluated by: (1) documented governance roles, contributor pathways, and software architecture, assessed by whether contributors outside the core team can act on them without direct consultation; (2) educational and onboarding materials, assessed through pilot participant performance and observed barriers; (3) a framework for developing, reviewing, distributing, and maintaining community plug-ins, assessed through limited pilots of the proposed plug-in development framework; (4) a documented validation and benchmarking methodology, assessed by whether an independent laboratory can reproduce our accuracy checks on its own hardware and recordings; and (5) a sustainability model, assessed by review of its revenue and maintenance assumptions against the demand evidence gathered during discovery. Each output will be revised in response to stakeholder feedback before incorporation into the final ecosystem plan.

Qtr	Milestone / Deliverable	Evaluation
Q1–Q2	Complete I-Corps for PESOSE; establish baseline ecosystem metrics	Completion confirmed by I-Corps instructors
Q1–Q2	Stakeholder discovery; prioritized user and contributor barrier map	Target groups covered; recurring barriers confirmed
Q2	Develop candidate governance, contributor, and stewardship models	Compared with discovery findings and reviewed by OSE maintainers
Q2–Q3	Define extension contribution, maintenance, licensing, and review requirements	Tested against discovered extension needs; external OSE review
Q3	Security/data-quality scoping; prototype benchmarking and regression suite	Known output differences detected; risks mapped to controls
Q3	Pilot education and onboarding with partner organizations	Participant feedback and observed barriers
Q4	Integrate findings into governance, sustainability, community, and implementation roadmap	Processes usable without direct core-team consultation

Broader Impacts

In a fully developed ecosystem, the same software can serve a thirteen-year-old with a couple of webcams and a federally funded research laboratory studying human perceptuomotor neuroscience. A contribution made by an animator can improve the software for researchers and clinicians, and scientific advancements can become accessible to artists, educators, students, and hobbyists. A sustainable FreeMoCap ecosystem can turn specialized adaptations into shared research infrastructure, increasing the return on public investment while accelerating the translation of research advances across fields and into practice.

The educational activities piloted under this proposal will help develop and test materials for users of differing technical backgrounds to progress towards independent software use, and where desired, contribution. A mature ecosystem can teach a user how a pixel in an image becomes a 3D point in space, not just how to run the software that does it. By pairing extensible infrastructure with education and cross-domain participation, FreeMoCap, supported by a developed ecosystem, can broaden not only who has access to quantitative measurement tools, but also who is able to understand, adapt, and help shape them.

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SUMMARY PROPOSAL BUDGET

YEAR 1

ORGANIZATION FREEMOCAP FOUNDATION, INC				FOR NSF USE ONLY		
				PROPOSAL NO.	DURATION (months)	
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Jonathan Matthis				Proposed		Granted
				AWARD NO.		
A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer
				CAL	ACAD	SUMR
1. Jonathan Matthis - Principal Inv				8.0		64,250
2. Aaron Cherian				9.0		64,250
3.						
4.						
5.						
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.0		0
7. (2) TOTAL SENIOR/KEY PERSONNEL (1 - 6)				17.0		128,500
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)						
1. (0) POST DOCTORAL SCHOLARS				0.0		0
2. (1) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				9.0		44,250
3. (0) GRADUATE STUDENTS						0
4. (0) UNDERGRADUATE STUDENTS						0
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)						0
6. (0) OTHER						0
TOTAL SALARIES AND WAGES (A + B)						172,750
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)						34,550
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)						207,300
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$10,000.)						
TOTAL EQUIPMENT						0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)						10,000
2. INTERNATIONAL						0
F. PARTICIPANT SUPPORT COSTS						
1. STIPENDS \$ _____ 0						
2. TRAVEL _____ 0						
3. SUBSISTENCE _____ 0						
4. OTHER _____ 0						
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS						0
G. OTHER DIRECT COSTS						
1. MATERIALS AND SUPPLIES						8,000
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION						0
3. CONSULTANT SERVICES						32,000
4. COMPUTER SERVICES						0
5. SUBAWARDS						0
6. OTHER						2,500
TOTAL OTHER DIRECT COSTS						42,500
H. TOTAL DIRECT COSTS (A THROUGH G)						259,800
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) Indirect (Rate: 15.0, Base:259800)						
TOTAL INDIRECT COSTS (F&A)						38,970
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)						298,770
K. FEE						0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)						298,770
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$		
PI/PD NAME Jonathan Matthis				FOR NSF USE ONLY		
ORG. REP. NAME*				INDIRECT COST RATE VERIFICATION		
				Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

SUMMARY PROPOSAL BUDGET

Cumulative

ORGANIZATION FREEMOCAP FOUNDATION, INC				FOR NSF USE ONLY		
				PROPOSAL NO.	DURATION (months)	
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Jonathan Matthis				Proposed		Granted
				AWARD NO.		
A. SENIOR/KEY PERSONNEL: PI/PD, Co-PI's, Faculty and Other Senior/ Key Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer
				CAL	ACAD	SUMR
1. Jonathan Matthis - Principal Inv				8.0		64,250
2. Aaron Cherian				9.0		64,250
3.						
4.						
5.						
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)						
7. (2) TOTAL SENIOR/KEY PERSONNEL (1 - 6)				17.0		128,500
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)						
1. (0) POST DOCTORAL SCHOLARS				0.0		0
2. (1) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				9.0		44,250
3. (0) GRADUATE STUDENTS						0
4. (0) UNDERGRADUATE STUDENTS						0
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)						0
6. (0) OTHER						0
TOTAL SALARIES AND WAGES (A + B)						172,750
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)						34,550
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)						207,300
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$10,000.)						
TOTAL EQUIPMENT						0
E. TRAVEL 1. DOMESTIC (INCL. U.S. POSSESSIONS)						10,000
2. INTERNATIONAL						0
F. PARTICIPANT SUPPORT COSTS						
1. STIPENDS \$ <u>0</u>						0
2. TRAVEL <u>0</u>						0
3. SUBSISTENCE <u>0</u>						0
4. OTHER <u>0</u>						0
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS						0
G. OTHER DIRECT COSTS						
1. MATERIALS AND SUPPLIES						8,000
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION						0
3. CONSULTANT SERVICES						32,000
4. COMPUTER SERVICES						0
5. SUBAWARDS						0
6. OTHER						2,500
TOTAL OTHER DIRECT COSTS						42,500
H. TOTAL DIRECT COSTS (A THROUGH G)						259,800
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)						
TOTAL INDIRECT COSTS (F&A)						38,970
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)						298,770
K. FEE						0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)						298,770
M. COST SHARING PROPOSED LEVEL \$ 0				AGREED LEVEL IF DIFFERENT \$		
PI/PD NAME Jonathan Matthis				FOR NSF USE ONLY		
ORG. REP. NAME*				INDIRECT COST RATE VERIFICATION		
				Date Checked	Date Of Rate Sheet	Initials - ORG

*ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

Budget Justification

Senior/Key Personnel (Line A)

Salaries for two senior/key personnel are requested. All personnel budgeted on Lines A and B will be employees of the FreeMoCap Foundation, the proposing organization, and will be acting as its employees while performing the responsibilities described below. NSF does not recognize executive titles for salary-rate determination; the rates below therefore correspond to each individual's functional responsibilities, and each is well below the 75th-percentile Bureau of Labor Statistics rate for the corresponding occupation in the geographic area where the work will be carried out. Standard occupational classification codes and live BLS references are listed per person. Person-month calculations use 173.33 hours per month.

The two senior/key personnel are budgeted at different rates because they occupy different occupations, not different ranks. The PI's responsibilities classify under Computer and Information Research Scientists (SOC 15-1221) and the co-PI's under Software Developers (SOC 15-1252). Those occupations carry different BLS wage distributions, and each rate is benchmarked against its own occupation rather than against the other person's.

Jonathan Samir Matthis — Principal Investigator

- **Functional title / SOC:** Computer and Information Research Scientist (SOC 15-1221); BLS OEWS reference: bls.gov/oes/current/oes151221.htm
- **Salary rate:** \$96,375/year; **Effort:** 8.0 calendar months; **Requested:** \$64,250
- **Responsibilities:** Overall scientific and organizational direction of the project. Leads governance scoping (comparison and evaluation of stewardship models), the risk-analysis and security plan, and the core-versus-extension architecture decisions. Serves as Technical Lead for the mandatory I-Corps for PESOSE training, with decision-making authority over project direction, including approximately one day per week of ecosystem-discovery interviewing across the training window. The remaining 4 months of his effort continue on unfunded Foundation activities.

Aaron Cherian — co-Principal Investigator

- **Functional title / SOC:** Software Developer (SOC 15-1252); BLS OEWS reference: bls.gov/oes/current/oes151252.htm
- **Salary rate:** \$85,667/year; **Effort:** 9.0 calendar months; **Requested:** \$64,250
- **Responsibilities:** Leads the technical scoping work: definition of the community-extension architecture, design of the validation, benchmarking, and regression-testing infrastructure, and coordination of the data-quality and provenance metadata work. Serves as Entrepreneurial Lead for the I-Corps for PESOSE training, coordinating stakeholder outreach, discovery interviews, and weekly reporting, at approximately one day per week across the training window. The remaining 3 months of his effort continue on unfunded Foundation activities.

Salary support in excess of two months

The PESOSE program does not enforce the general two-month limit on senior personnel salary support, but requests exceeding it require justification under PAPPG II.D.2.f(i)(a). The Foundation defines “year” as the 12-month project period of this award, and applies that definition consistently across all personnel and all NSF proposals.

The requested effort exceeds two months because the employment situation differs from the academic case the policy addresses. NSF's premise is that research is a normal function of faculty and that compensation for it is already included in an institutional salary. Neither individual holds an academic appointment or any other salaried position from which this effort could be drawn,

and this award is the sole source of compensation for the work described. The requested salary is not supplemental to an existing salary; it is the compensation for the work itself.

The scope of effort is also set by the program. PESOSE requires a minimum of 100 ecosystem-discovery interviews and approximately one day per week of interviewing from both the Technical Lead and the Entrepreneurial Lead, plus kick-off, weekly sessions, and a closing meeting. The PESOSE budget guidance directs proposers to budget salary support sufficient to accommodate it.

Other Personnel (Line B)

Paul Matthias — Research Infrastructure Engineer

- **Functional title / SOC:** Network and Computer Systems Administrator (SOC 15-1244); BLS OEWS reference: bls.gov/oes/current/oes151244.htm
- **Salary rate:** \$59,000/year; **Effort:** 9.0 calendar months; **Requested:** \$44,250
- **Responsibilities:** Builds and maintains the technical infrastructure on which the planning activities depend: release and continuous-integration infrastructure (GitHub Actions), project web services, and the Google Cloud Platform environment hosting project telemetry, including the provenance and data-quality instrumentation scoped in the Risk Analysis and Security Plan. Implements the technical environment for the community education pilots, and authors and maintains the contributor-facing operational documentation for that infrastructure. This is a technical position; no administrative or clerical effort is direct-charged to this award.

Fringe Benefits (Line C)

Fringe benefits are requested at a rate of 20% applied to a base of \$172,750, comprising all salaries requested on Lines A and B. Requested fringe: \$34,550. The rate is composed as follows:

Component	Rate	Basis
Statutory payroll taxes (FICA: Social Security and Medicare)	7.65%	Federal statutory rate
Employer contribution toward individual health insurance coverage	9.35%	Estimated employer contribution
Employer retirement plan contribution	3.00%	Estimated employer contribution
Total fringe rate	20.00%	

Beyond the statutory payroll tax component, the rates above are estimates. The Foundation is standing up employer-of-record, payroll, and benefits administration ahead of the period of performance, and the health and retirement contribution levels will be set in that work; that administrative capacity is funded through indirect cost recovery rather than charged directly to this award. The 20% composite rate applies uniformly to all project personnel. The award covers a single 12-month period; no salary escalation is applied or requested.

Equipment (Line D)

No equipment is requested. Funding requests for equipment are not permitted for PESOSE Track 1 proposals.

Travel (Line E)

Total travel requested is \$10,000, all domestic: \$7,000 for two conference trips supporting community building and dissemination, and \$3,000 for I-Corps ecosystem-discovery travel. No international travel is requested; the program does not permit it. Costs are budgeted under 2 CFR § 200.475.

Trip 1 — American Society of Biomechanics (ASB) annual meeting, 1 traveler (PI). Engagement with the biomechanics research community that constitutes FreeMoCap’s core academic user base; dissemination of results and recruitment of contributors and ecosystem partners.

Trip 2 — Society for Neuroscience (SfN) annual meeting, 1 traveler (co-PI). Engagement with the neuroscience research community, a major user population for accessible motion measurement, and a venue for ecosystem-discovery interviews with prospective users and contributors.

Cost element	Trip 1 (ASB)	Trip 2 (SfN)	Total
Airfare (domestic, round trip)	\$700	\$750	\$1,450
Lodging (4 nights)	\$1,100	\$1,200	\$2,300
Meals and incidentals (5 days)	\$425	\$425	\$850
Conference registration	\$900	\$750	\$1,650
Ground transportation	\$375	\$375	\$750
Trip total	\$3,500	\$3,500	\$7,000

No booth, exhibiting, or other marketing-related expenses are requested.

Ecosystem-discovery travel (I-Corps for PESOSE) — \$3,000. Four domestic trips at approximately \$750 each (airfare or ground transportation, one night lodging, meals and incidentals) to conduct in-person interviews during the training window. The training requires a minimum of 100 discovery interviews, and NSF’s I-Corps guidance expressly permits domestic travel to conduct them in person.

Participant Support (Line F)

No participant support costs are requested. The education and onboarding pilots are conducted with partner organizations that host their own learners; no stipends, subsistence, travel allowances, or registration fees are paid to or on behalf of pilot participants. I-Corps discovery interviewees are likewise uncompensated, consistent with the program’s prohibition on incentives for interviewees.

Other Direct Costs (Line G)

Materials and Supplies — \$8,000

Multi-camera capture kits (consumer-grade webcams and mounting hardware) and workshop materials for the hands-on education pilots with partner organizations, and for hardware-configuration testing across supported platforms. All are consumable or low-cost items; no single item approaches the \$5,000 per-unit federal threshold for equipment. This request is under 10% of the total budget.

I-Corps for PESOSE — \$8,500 total across budget lines

Participation in the mandatory I-Corps for PESOSE training is budgeted at \$8,500, within the \$30,000 program ceiling. Its components fall on three different budget lines and are gathered here for review:

- I-Corps participation fee: \$2,500 — this line (Other Direct Costs)
- Industry Mentor stipend: \$3,000 — Consultant Services, below
- Ecosystem-discovery travel within the training window: \$3,000 — Travel (Line E), above

Salary support for the Technical and Entrepreneurial Leads’ I-Corps effort is carried on Line A and excluded from the \$8,500 above to avoid double-counting. The training team is the Technical Lead (Jonathan Samir Matthis), Entrepreneurial Lead (Cherian), and Industry Mentor (Idehen)

— three individuals, meeting the requirement that Track 1 teams budget for at least three and no more than five. No gifts, meals, marketing expenses, or survey tools are requested.

Consultant Services — \$32,000

Track 1 is a scoping and planning award, and several of the mandated planning areas call for independent professional expertise the project team does not itself hold. Consultants are engaged to produce assessments, evaluations, and plans — not to develop the open-source product, which is outside the scope of a Track 1 award. None is an officer or employee of the proposing organization. Expertise, primary organizational affiliation, normal daily compensation rate, and number of days of expected service are given for each below, per PAPPG II.D.2.g(v)(c). No consultant will receive more than \$50,000 over the award period. The Foundation is a nonprofit organization; no owners or equity holders exist whose compensation would be restricted.

Consultant / expertise	Daily rate	Days	Total
Open-source governance and licensing counsel	\$1,800	10.0	\$18,000
Independent security and supply-chain assessment	\$2,000	4.5	\$9,000
Community documentation and education-pilot facilitation	\$500	4.0	\$2,000
I-Corps Industry Mentor (Endurance Idehen)	\$500	6.0	\$3,000
Total		24.5	\$32,000

Legal services are budgeted here rather than as a separate line because NSF treats professional service costs and consultant services as the same category. The legal work requested is specific to the project’s scope of work — the licensing and governance analysis the solicitation requires Track 1 proposals to produce. General organizational legal costs, including incorporation, nonprofit compliance, board governance, and employment law, are treated as general administration, are recovered through the indirect cost rate, and are not direct-charged to this award.

Open-source governance and licensing counsel — \$18,000. Independent legal and policy expertise supporting the Organization and Governance planning area, which the solicitation requires Track 1 proposals to address: comparative evaluation of stewardship and fiscal-host models, licensing options for the core product and community-contributed extensions, trademark and project-naming policy, and contributor licensing terms. Deliverable is a written governance and licensing options analysis feeding the ecosystem’s organizational design.

Independent security and supply-chain assessment — \$9,000. External security expertise supporting the Risk Analysis and Security Plan, which the solicitation requires Track 1 proposals to address. Scope: threat modeling, review of the dependency graph and software-bill-of-materials posture, evaluation of secure release and chain-of-custody practices against Open Source Security Foundation and CISA/NSA supply chain guidance, and identity and access management review. Deliverable is a written vulnerability and risk assessment with prioritized remediation planning, produced independently of the development team.

Community documentation and education-pilot facilitation — \$2,000. Compensated engagement of experienced community members, budgeted across several individuals, for work that the Community Building planning area depends on: authoring and reviewing onboarding and contributor documentation, facilitating the education and onboarding pilots with partner organizations, and structured evaluation of the contributor experience. This funds documentation, facilitation, and assessment — not product feature development. The balance of the work is carried by the Research Infrastructure Engineer on Line B, whose effort was increased for that purpose.

I-Corps Industry Mentor — \$3,000. Endurance Idehen, an experienced open-source ecosystem leader, serves as Industry Mentor on the I-Corps for PESOSE training team. NSF’s I-Corps guidance requires a few hours per week across the training window in addition to program meetings; the request reflects 6.0 days at \$500/day. He is not an employee of the Foundation and is therefore budgeted here rather than on Line A or B.

Other — \$2,500

The I-Corps for PESOSE participation fee, described above.

Employer-of-record, payroll, and benefits administration services are not requested as direct costs. Those are general administrative functions of the organization, recovered through the indirect cost rate below, and consistent treatment under 2 CFR § 200.414(f) prohibits charging them both directly and through the de minimis rate.

No subawards are requested.

Indirect Costs (Line I)

The FreeMoCap Foundation does not have a federally negotiated indirect cost rate agreement and elects the de minimis rate of 15% of modified total direct costs, as permitted by the PESOSE budget guidance and 2 CFR § 200.414(f).

Modified total direct costs are \$259,800, equal to total direct costs. No equipment, participant support, tuition remission, or subawards are requested, so none of the MTDC exclusions in 2 CFR § 200.1 apply to this budget. Indirect costs: \$38,970.

Budget Summary

Line	Amount
A. Senior/Key Personnel (2 people, 17.0 cal months)	\$128,500
B. Other Personnel (1 person, 9.0 cal months)	\$44,250
C. Fringe Benefits (20% of \$172,750)	\$34,550
D. Equipment	\$0
E. Travel (domestic; international \$0)	\$10,000
F. Participant Support	\$0
G.1 Materials and Supplies	\$8,000
G.3 Consultant Services	\$32,000
G.6 Other (I-Corps participation fee)	\$2,500
G. Total Other Direct Costs	\$42,500
H. Total Direct Costs	\$259,800
I. Indirect Costs (15% of \$259,800 MTDC)	\$38,970
J. Total Requested	\$298,770

The total request of \$298,770 is within the \$300,000 Track 1 ceiling for a 12-month period of performance.

Facilities, Equipment and Other Resources

Organizational Resources

The FreeMoCap Foundation is an incorporated 501(c)(3) nonprofit organization serving as the long-term steward of the FreeMoCap project and its component repositories. The Foundation operates as a distributed, remote-first organization: core team members work from home offices, meet regularly in person in the Boston area, and attend scientific conferences relevant to movement research. The Foundation maintains the project’s public web presence at freemocap.org, hosted on a Foundation-managed server.

Community and Communications Infrastructure

The ecosystem’s principal resource is a community platform that already operates at scale:

- A GitHub organization hosting the project’s modular repository suite, with continuous integration and release automation handled through GitHub Actions;
- A Discord server of approximately 4,000 members, structured so that users can ask and answer questions without direct involvement of the core development team;
- A weekly community call providing a regular venue for project updates, user feedback, and contributor onboarding; and
- A tiered communications model that provides venues at three levels of openness: a private core-maintainer channel for internal coordination, the open Discord community for user support and discussion, and fully public channels — a YouTube channel, Twitch live streams, and social media accounts on Twitter and Bluesky — for announcements and public-facing content.

This layered communications structure is itself a resource for the proposed work: it provides the existing venues through which ecosystem discovery, community education, and contributor recruitment activities will operate.

Partner and Collaborator Resources

The Foundation works with Boston-area community organizations that provide physical space and learning communities for the education pilots described in this proposal, including The Possible Zone and Artisan’s Asylum. Workspace in Boston is additionally available to the team through the Boston Public Library’s nonprofit partnership program. Finally, the project draws on a distributed network of academic collaborators who make available domain expertise, populations, and settings for testing and validation across application areas.

Computing and Hardware

Core team members maintain desktop and laptop systems covering all supported platforms — Windows, Linux, and macOS on both Intel and Apple Silicon — sufficient to test releases across the full platform matrix. This aligns with the software’s design as motion capture for consumer-grade hardware: because the workflow runs on ordinary webcams, no specialized laboratory equipment is required to develop or validate it. Web hosting and telemetry services run on Google Cloud Platform space provided in-kind through the Google for Nonprofits program.

Data Management and Sharing Plan

Submitted to:

Dir for Tech, Innovation, & Partnerships (TIP), Translational Impacts (TI), POSE

1. FreeMoCap software, component repositories, and governance artifacts

Data or Research Product Category General Information

Data or Research Product Category: **FreeMoCap software, component repositories, and governance artifacts**

Description: **The existing FreeMoCap product and its component repositories, together with the governance, documentation, and contributor onboarding artifacts developed during the project.**

Access Policies and Limitations

Not applicable

Data Standards and Metadata

Data/Metadata Standard: **Semantic versioning with tagged, documented releases.**

Data or Research Product Provenance

Data Source: **New Data Collection**

Public Archiving

Data Sharing Location: **Public GitHub repositories, with tagged releases archived via the Zenodo–GitHub integration.**

Additional Details: **Each tagged release yields a citable DOI, so publications can reference immutable, permanently available versions. The distributed development model adds resilience: complete copies of the product exist across many independent stakeholders and hosting locations.**

Timeline for Public Accessibility

Data Sharing Timeline: **At time of collection**

Data Availability

Data Availability Period: **Indefinitely**

Will you adhere to the standard retention policy of the chosen repository: **Yes**

Accountability

Responsible Personnel: **Jonathan Matthis**

Role: **Principal Investigator**

Organization Associated within the Proposal: **FREEMOCAP FOUNDATION, INC**

Email Address: **jon@freemocap.org**

2. Ecosystem-discovery findings and aggregate usage telemetry

Data or Research Product Category General Information

Data or Research Product Category: **Ecosystem-discovery findings and aggregate usage telemetry**

Description: **Stakeholder findings, user and contributor barrier maps, and requirements documents from Track 1 discovery activities; and basic usage events keyed to a random, non-identifying per-installation UUID used to produce aggregate usage statistics.**

Access Policies and Limitations

Access Limitation(s)/Exception(s):

1. Human Data Protection

Explanation(s) and Justification(s) of Concern(s): - **Human-participant confidentiality** — raw interview recordings and transcripts from ecosystem-discovery activities will not be shared; only aggregated findings will be published. - **Personal privacy** — raw telemetry logs are retained internally and not shared; only aggregates will be published.

Data Standards and Metadata

Data/Metadata Standard: **Formats and schemas documented and versioned publicly where no community standard exists.**

Data or Research Product Provenance

Data Source: **New Data Collection**

Public Archiving

Data Sharing Location: **Public project repositories (aggregated findings only).**

Additional Details: **Aggregated discovery findings** — barrier maps, requirements documents, and summarized stakeholder results — are published in the project's public GitHub repositories alongside the governance and documentation artifacts, and any findings supporting a publication are additionally deposited in a DOI-issuing public repository at the time of publication, per NSF public access requirements. Aggregate usage statistics derived from telemetry are published on the same cadence. The withheld source material in this category — raw interview recordings and transcripts, and raw telemetry event logs — is retained on Foundation-controlled internal storage and is not deposited in any public repository.

Timeline for Public Accessibility

Data Sharing Timeline: **At time of publication**

Data Availability

Data Availability Period: **Indefinitely**

Will you adhere to the standard retention policy of the chosen repository: **Yes**

Accountability

Responsible Personnel: **Jonathan Matthis**

Role: **Principal Investigator**

Organization Associated within the Proposal: **FREEMOCAP FOUNDATION, INC**

Email Address: **jon@freemocap.org**

3. Validation and benchmark datasets

Data or Research Product Category General Information

Data or Research Product Category: **Validation and benchmark datasets**

Description: **Reference recordings and cross-version regression baselines supporting publications and ongoing benchmarking.**

Access Policies and Limitations

Access Limitation(s)/Exception(s):

1. Human Data Protection

Explanation(s) and Justification(s) of Concern(s): **Identifiable raw source video from validation recordings is not released. Deposited datasets comprise de-identified annotated video and derived kinematic data; the documented folder structure separates these from raw source video so that derived data can be released without it. Media contributed voluntarily by community members is released only within the consent tier the contributor selected at submission.**

Data Standards and Metadata

Data/Metadata Standard: **Documented, versioned folder structure separating raw video, de-identified annotated video, and derived kinematic data; provenance metadata sufficient to trace any dataset to the software versions, pose-estimation models, and processing parameters that generated it**

Data or Research Product Provenance

Data Source: **New Data Collection**

Public Archiving

Data Sharing Location: **DOI-issuing public repository, deposited at time of publication per NSF public access requirements**

Additional Details: **Reference recordings and cross-version regression baselines are deposited in a DOI-issuing public repository at the time of the associated publication, per NSF public access requirements, with tagged software releases archived in parallel through the Zenodo-GitHub integration so that every dataset can be cited alongside the exact software version, pose-estimation model, and processing parameters that produced it. Datasets follow the project's documented, versioned folder structure separating raw video, de-identified annotated video, and derived kinematic data, so that a depositor can release derived data without releasing identifiable source video. Archived material in this category is limited to recordings the project collects and curates deliberately for validation purposes. FreeMoCap is a standalone desktop application in which all processing occurs on the user's own computer; recordings of identifiable human participants never leave the user's machine, and the workflow is operable in fully air-gapped settings. The ecosystem archives and shares software, derived kinematic data, and benchmarks, and is not a repository of user biospecimens. Media contributed voluntarily by community members is governed by tiered, per-submission consent covering both identifiability and sharing context, and is redistributed only within the tier the contributor selected.**

Timeline for Public Accessibility

Data Sharing Timeline: **At time of publication**

Data Availability

Data Availability Period: **Indefinitely**

Will you adhere to the standard retention policy of the chosen repository: **Yes**

Accountability

Responsible Personnel: **Jonathan Matthis**

Role: **Principal Investigator**

Organization Associated within the Proposal: **FREEMOCAP FOUNDATION, INC**

Email Address: **jon@freemocap.org**

4. Educational and training materials

Data or Research Product Category General Information

Data or Research Product Category: **Educational and training materials**

Description: **Introductory, intermediate, and developer-facing materials developed and piloted with community partners.**

Access Policies and Limitations

Not applicable

Data Standards and Metadata

Data/Metadata Standard: **No community metadata standard exists for educational materials of this kind. Materials are authored in open, plain-text formats (Markdown, Jupyter notebooks) with accompanying media, carry front-matter identifying audience level, prerequisites, and the software version they were written against, and follow the project's per-repository layout convention so that documentation sits in the same location in every repository.**

Data or Research Product Provenance

Data Source: **New Data Collection**

Public Archiving

Data Sharing Location: **Public GitHub repositories, versioned alongside the software they document.**

Additional Details: **Introductory, intermediate, and developer-facing materials are published in the project's public GitHub repositories as they are developed, and are versioned alongside the software release they were written against so that a learner can retrieve the materials matching the version they are running. Materials piloted with partner organizations are published on the same basis as those developed internally. All materials remain publicly available indefinitely.**

Timeline for Public Accessibility

Data Sharing Timeline: **Released on a rolling basis as materials are developed and piloted.**

Data Availability

Data Availability Period: **Indefinitely**

Will you adhere to the standard retention policy of the chosen repository: **Yes**

Accountability

Responsible Personnel: **Jonathan Matthis**

Role: **Principal Investigator**

Organization Associated within the Proposal: **FREEMOCAP FOUNDATION, INC**

Email Address: **jon@freemocap.org**

List of Suggested Reviewers

Data Not Available

List of Reviewers Not to Include

Data Not Available

Other Personnel Biographical Information

Data Not Available

SAN JOSÉ STATE UNIVERSITY

Digital Media Art · CADRE Laboratory for New Media
One Washington Square, San José, CA 95192

August 30, 2026

National Science Foundation
Pathways to Enable Secure Open-Source Ecosystems (PESOSE) Program

Re: Letter of Collaboration for the FreeMoCap PESOSE Proposal

To Whom It May Concern:

I am writing regarding our ongoing use of and collaboration with the FreeMoCap project. I am a senior lecturer with the CADRE Laboratory for New Media at San José State University, where I teach digital media art with a focus on 3D tools and animation, and where both my teaching and my studio practice run on free and open-source software.

We use FreeMoCap in two connected ways. In the classroom, it gives students a way to work directly with motion capture on accessible hardware and open-source software, and to see the whole process of recording, processing, and applying motion data rather than treating it as a black box. In my own practice, FreeMoCap is the engine of a six-camera capture studio I built from secondhand webcams at BASEMENT, an artist-run space in San Francisco, where it drives the character animation for my current film work.

That use has led us to build tools around the FreeMoCap workflow. The main one is HIHO Mocap, a free and open-source Blender add-on designed to work with FreeMoCap output and support the animation and teaching workflows we use at CADRE. Developing it has given us practical experience with how FreeMoCap can be extended for artists, students, and classrooms.

Working with students has also surfaced the practical questions that come with bringing motion capture into education: what hardware makes a reliable student setup, how a system can be packaged into a reproducible classroom kit, what documentation actually helps learners at different levels of technical experience, and how complex processing workflows can be presented clearly. The capture pipeline is now installed in our classroom at SJSU, and our students this fall are its first testers.

If the proposed PESOSE project is funded, we would be glad to keep collaborating with the FreeMoCap team on these educational and creative fronts. That could include feedback from classroom use, help evaluating documentation and student-facing workflows, exploring FreeMoCap-based kits or standardized setups for teaching environments, and continuing to develop and test open tools that connect FreeMoCap with Blender and related digital-media workflows.

We look forward to continuing to share what we learn as educators, artists, and users of FreeMoCap as these tools grow.

Sincerely,

David Bayus

Senior Lecturer, Digital Media Art
CADRE Laboratory for New Media, San José State University
david.bayus@sjsu.edu · www.davidbayus.zone

Relationship to proposing team: External FreeMoCap user, contributor, and educational collaborator



We envision a world where young people shape their own futures.

August 8th, 2026

National Science Foundation
Pathways to Enable Secure Open-Source Ecosystems (PESOSE) Program

Re: Letter of Collaboration for the FreeMoCap PESOSE Proposal

To Whom It May Concern:

My name is Raven Tukes, Chief Partnerships Officer at The Possible Zone (TPZ). I am writing on behalf of (TPZ) to express our interest in collaborating with the FreeMoCap Foundation as part of its proposed NSF PESOSE project.

The Possible Zone is a Boston-based nonprofit educational organization that provides young people with hands-on, experiential learning opportunities that build entrepreneurial spirit, technical skills, and pathways toward future education and careers. Through our STEAM-focused programming and Innovation Center, our students engage directly with emerging technologies while developing the skills and confidence necessary to explore how those technologies can be used, adapted, and created.

FreeMoCap's mission to make advanced human-movement measurement accessible through free and open-source software offers a hands-on way for students to explore technologies that combine computer vision, artificial intelligence, biomechanics, video, and data analysis. These activities can make technical concepts that might otherwise remain abstract more tangible and accessible.

If the proposed project is funded, The Possible Zone intends to collaborate with the FreeMoCap team to develop and pilot educational activities that introduce students to motion capture, computer vision, biomechanics, and open-source software through hands-on use of the FreeMoCap platform. As an initial activity, we anticipate working together to develop and deliver a workshop for TPZ students, using the experience to identify which concepts and activities are most engaging and accessible to learners with a range of technical backgrounds, as well as barriers students encounter when first engaging with the software.

Feedback from students and TPZ educators would inform subsequent refinement of these educational materials. Based on the outcomes of this initial work and mutual interest, TPZ and FreeMoCap may also explore opportunities to develop the material into a more substantial educational experience within TPZ's existing programming

model, such as a STEAM Deep Dive or Career Exploration LaunchLab. This could provide students with an opportunity not only to use an emerging technology, but also to explore the computer vision, artificial intelligence, data analysis, and open-source development concepts that make the technology possible.

Through this collaboration, TPZ would help the FreeMoCap team develop and evaluate educational approaches for lowering barriers to participation in an open-source scientific software community. Lessons learned through the workshop and any subsequent programming would inform educational resources intended for broader use by educators, students, and other new members of the FreeMoCap community.

Sincerely,

A handwritten signature in black ink, appearing to read "Raven Tukes".

Raven Tukes

Chief Partnerships Officer

The Possible Zone

Relationship to proposing team: External educational collaborator



Letter of Collaboration

26 August 2026

To the National Science Foundation,

I am writing on behalf of Imatest LLC to express our interest in collaborating with the FreeMoCap Foundation if it is selected for funding through the NSF Pathways to Enable Secure Open-Source Ecosystems (PESOSE) program.

Imatest develops software and test methods for measuring and characterizing the performance of cameras and imaging systems. Our work focuses on enabling designers and users of imaging systems to understand how camera hardware, image processing, and other parts of the imaging pipeline affect the quality and information content of captured images. In addition to my work at Imatest, I am involved in the ongoing ISO/TC 42/WG 18 work related to image-information content (ISO/WD 23654).

There is a natural overlap between this work and some of the challenges FreeMoCap faces as an open source markerless motion capture system. FreeMoCap is intentionally hardware-agnostic and can be used with relatively inexpensive consumer cameras. This raises important questions about what makes a camera suitable for computer-vision-based motion measurement and how those characteristics should be measured and reported. Users of FreeMoCap need to be able to distinguish limitations introduced by the imaging system from limitations introduced later in the computer-vision and processing pipeline.

If funded, we would be interested in continuing discussions with the FreeMoCap team around these questions and exploring areas where Imatest's experience with image and video quality measurement may be useful. This could include sharing our perspective on relevant spatial and temporal imaging characteristics, camera characterization methods, and applicable standards, as well as exploring how camera characterization might inform areas such as hardware recommendations or benchmarking practices. The nature and scope of the collaboration would depend on the needs of the project and areas of mutual interest.

My involvement in ongoing standards work related to image-information content may also provide a useful perspective for these discussions. The questions that FreeMoCap is asking about camera suitability and measurement quality have clear connections to broader questions about how the information produced by imaging systems can be characterized and evaluated. FreeMoCap may also provide an interesting real-world context for considering some of the practical challenges that arise when these concepts are applied to motion capture.

We see value in bringing an imaging-system perspective to FreeMoCap's broader work. Considering the role of camera hardware alongside the computer-vision and processing pipeline may help users and contributors better understand the factors that affect motion-capture measurements and make more informed decisions about cameras, acquisition setups, and downstream analysis.

Sincerely,

Sarah Kerr
Imaging Scientist
Imatest LLC

Relationship to proposing team: External technical collaborator



Department of Physiology
and Biophysics

SCHOOL OF MEDICINE

UNIVERSITY OF COLORADO

ANSCHUTZ MEDICAL CAMPUS

Benjamin Scholl

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Aurora, CO 80045, USA

www.medschool.cuanschutz.edu/physiology

Re: NSF Pathways to Enable Secure Open-Source Ecosystems (PESOSE) Program Application

To Whom It May Concern:

I am writing on behalf of Dr. Jonathan Matthis, with whom I have had an ongoing collaboration with through the FreeMoCap Foundation since 2024.

Our lab studies the development of natural visual and sensorimotor behavior in the freely moving ferret. Specifically, we are focused on the rapid development of oculomotor skills after eye opening. Our recording system and analysis pipeline **required simultaneous measurements of natural behavior and neural activity**, including binocular eye movements, three-dimensional head and body movement, and recordings from visual cortex. Because animals are developing, markerless motion-capture is the only available method and standard human workflows would not work. Thus, in a collaboration with the FreeMoCap team, a 3D reconstruction and analysis pipeline using multiple camera systems was built. **Without FreeMoCap our work would not be possible.**

One component of this pipeline is SkellyClicker, an open-source tool that makes custom DeepLabCut tracking models easier to create and use within motion-capture workflows. The FreeMoCap team led its development, with individual member of my lab contributing to design, testing, and day-to-day usability. Because SkellyClicker supports custom tracking models, the same workflow can be adapted for other researchers working with nonhuman animals or other tracking problems that are not well served by standard human pose-estimation models.

With funding from PESOSE project, our collaboration with the FreeMoCap ecosystem can accelerate— as both scientific users and contributors. Our lab expects to continue contributing to the development and evaluation of tools and workflows for nonhuman tracking, while providing feedback on technical interfaces, documentation, and contributor processes that can make specialized extensions easier to develop and share. We are excited to continue this collaboration and know that through it, we are able to tackle leading scientific questions and develop novel unique research directions.

Sincerely,

Benjamin Scholl, Ph.D.

Assistant Professor

Department of Physiology and Biophysics

University of Colorado Anschutz Medical Campus

Letter of Collaboration | External FreeMoCap Users and Contributors

31 August 2026

To the National Science Foundation,

This document compiles brief statements from external FreeMoCap users and contributors who have independently become involved with the project through animation, software development, interface design, and professional game production. Together, these statements describe their existing contributions and use of FreeMoCap, as well as their anticipated continued involvement in the FreeMoCap ecosystem.

Andres Casals

Animation / Blender

My name is Andres Casals, and I am a Telecommunications Engineer. I first became involved with FreeMoCap through a personal animation project, creating a music video in Unreal Engine. When I reached the character animation stage, I tried several free tiers of commercial motion capture software, but none produced usable results. After searching YouTube for open-source alternatives, I stumbled upon the FreeMoCap project. It not only provided the tools I needed for my project, but also gave me complete freedom to modify the software and contribute back to the community.

Initially, my contributions focused on the Blender animation export pipeline to support my own workflow goals: correcting marker orientation, applying rigid body logic to markers, generating an armature compatible with the Unreal Engine MetaHuman, implementing a foot-locking function, and improving the Blender add-on UI to streamline exporting animations to Unreal Engine.

Given my engineering background, the scientific aspects of FreeMoCap also naturally appealed to me. I contributed several visualization tools in Blender, including Base of Support, joint angles, and overlay charts for joint angles and COM position. Additionally, I have contributed to user onboarding and instructional materials by creating video tutorials, live demos, camera benchmarks, and testing new features.

I look forward to continuing my work with FreeMoCap, primarily focusing on the animation pipeline—an area I am passionate about. I strongly align with the open-source ethos of making motion capture tools as accessible as possible. My goal is to keep improving the software with user-friendly, animation-focused tools and well-documented workflows.

Andres Casals · Relationship to proposing team: External FreeMoCap contributor

Dominic Scott

Pose Estimation / Real-Time Analysis

I work with realtime motion capture tools, with a focus on making open source tools easier to deploy and use in real-world applications. Previously I created a motion capture plugin for a realtime graphics tool called TouchDesigner, and became involved in FreeMoCap when I saw the work being done to create a realtime pipeline as well. I wanted to contribute the knowledge I had learned from previous projects to help, and have since contributed across several parts of the technical pipeline.

My work with FreeMoCap has focused particularly on making pose-estimation models easier to deploy and on developing capabilities for real-time metrics and analysis. This has included specifying a standard for model metadata that allows for drag-and-drop inclusion of new models into FreeMoCap, and acceleration techniques to run models as quickly as possible across all platforms and hardware types. A recurring goal of this work has been reducing the technical overhead required to bring new models into practical use within FreeMoCap and expanding cross-platform compatibility.

Going forward, I hope to contribute further to the development of FreeMoCap's pose-estimation and real-time analysis capabilities. I am particularly interested in continuing to improve how new models can be integrated into the software and in helping develop workflows that make these capabilities easier to extend and use.

Dominic Scott · Relationship to proposing team: External FreeMoCap contributor

Pooya Moradi Motlagh

Product Design / UI/UX / Frontend Development

My name is Pooya Moradi M, and my background is in product design, UI/UX, and frontend development. I became involved with FreeMoCap as an external contributor.

My contributions have mainly focused on the user interface and overall usability of the software. This has included designing interfaces and user experiences in Figma, building frontend components using React and CSS, and creating a shared design system that provides a consistent foundation for the project and makes it easier for me and other contributors to work on the interface. A lot of this work has been about making the software feel clearer, more intuitive, and easier to use.

I plan to continue contributing to FreeMoCap as the software evolves, particularly through UI/UX design, frontend development, improving the design system, and making user-facing workflows easier to understand and use.

Pooya Moradi Motlagh · Rome, Italy · Product Designer · pooyamoradim@gmail.com · GitHub: @pooyadeperson 28 Aug 2026 · Relationship to proposing team: External FreeMoCap contributor

Jonathan Nilsson — GIBBET GAMES AB

Professional Game Development / Animation

I am writing on behalf of GIBBET GAMES AB regarding our use of and collaboration with the FreeMoCap project. GIBBET GAMES is an independent video game studio in Skövde, Sweden, working with game development, animation, rigging, motion capture, and technical animation.

We began using FreeMoCap during our work on *Esoteric Ebb*, where we used markerless motion capture extensively for character animation. FreeMoCap allowed us to record motion directly in our own workspace and iterate much faster than when relying on access to a dedicated OptiTrack studio.

We have since continued developing FreeMoCap-based workflows for game production. This includes transferring motion data through Blender into Autodesk Maya and Advanced Skeleton, as well as developing our own Maya tools for working with FreeMoCap animation, synchronized camera views, and recorded video.

Through our use of FreeMoCap, we have been in contact with members of the development team and have provided feedback from the perspective of game and animation production. We plan to continue using FreeMoCap in future projects and are interested in continuing to test, develop, and share animation-focused workflows with the FreeMoCap community.

If the proposed PESOSE project is funded, we would be happy to remain involved as an external user and collaborator and provide feedback from professional game-development and animation workflows.

Jonathan Nilsson · GIBBET GAMES AB · Relationship to proposing team: External FreeMoCap user and collaborator

List of Project Personnel, Collaborators, and Partner Organizations

Full Name	Organization(s)	Role in the Project
Jonathan Samir Matthis	FreeMoCap Foundation, Inc., Boston, MA	Principal Investigator
Aaron Cherian	FreeMoCap Foundation, Inc., Boston, MA	co-Principal Investigator
Paul Matthis	FreeMoCap Foundation, Inc., Boston, MA	Research Infrastructure Engineer (Other Personnel)
Endurance Idehen	Chorus Innovations, Austin, TX	I-Corps for PESOSE Industry Mentor (funded consultant)
David Bayus	San José State University, CADRE Laboratory for New Media, San José, CA	Collaborator; letter of collaboration
Jonathan Nilsson	GIBBET GAMES AB, Skövde, Sweden	Collaborator; letter of collaboration
Raven Tukes	The Possible Zone, Boston, MA	Collaborator; letter of collaboration
Sarah Kerr	Imatest LLC, Boulder, CO	Collaborator; letter of collaboration
Andres Casals	Independent, Rio de Janeiro, Brazil	Contributor; letter of collaboration
Dominic Scott	Independent, Toronto, Canada	Contributor; letter of collaboration
Pooya Moradi Motlagh	Independent, Rome, Italy	Contributor; letter of collaboration
Benjamin Scholl	University of Colorado Anschutz, Department of Physiology and Biophysics, Aurora, CO	Collaborator; letter of collaboration

Synergistic Activities — Jonathan Samir Matthis

1. **Development and open distribution of accessible research tooling.** Created and maintains FreeMoCap, a free, open-source markerless motion capture platform, and deliberately architected it as a suite of modular, independently documented repositories (camera synchronization, pose estimation, data processing, visualization) so that researchers can understand, deploy, and extend individual pipeline stages in their own projects rather than adopting an opaque monolith.
2. **Technical training and community support at scale.** Provides extensive informal training and technical support to a community of several thousand students, early-career researchers, and professionals — including guidance on conducting and publishing their work in independent, publicly visible ways — through community channels, written documentation, and regular live tutorial streams.
3. **Public, reproducible research-software development.** Conducts the entire development process “in public”: all work is hosted openly on GitHub with intentional documentation of both the technical pipeline and the organizational processes behind it, explicitly to make it easier for other groups to follow the same open, reproducible development model.
4. **Contributions to the broader open-source and research community.** Regularly contributes code review, pull requests, and technical guidance to third-party open-source projects, and provides public technical feedback to researchers presenting motion-capture and computer-vision methods work outside traditional review venues.
5. **Public science communication.** Produces publicly available video content (YouTube tutorials and Twitch live streams) on motion capture methods and open-source scientific software, and has discussed accessible research tooling and open science in podcast appearances.

Synergistic Activities — Aaron Cherian

1. **Open-source research tooling for data quality and benchmarking.** Designs and builds diagnostics and benchmarking tools for the FreeMoCap platform, giving the research community practical instruments for assessing the quality and reliability of markerless motion-capture data across systems and software versions.
2. **Validation research and publication.** Conducts validation studies of open-source markerless motion capture against established marker-based standards and publishes the results in the research literature, providing the community with openly available evidence about the capabilities and limitations of accessible motion-capture methods.
3. **Collaborative research across application domains.** Co-authors research papers with external researchers applying motion capture in their own domains, supporting the adoption of accessible movement measurement in fields far from the original development team's expertise and transferring methods knowledge to new research communities.
4. **Interface regularization and contributor onboarding.** Works to regularize and document the project's application programming interfaces and validation strategies, explicitly to lower the technical barriers for outside developers and researchers to contribute to the ecosystem — transferring architectural knowledge beyond the core development team.
5. **Community teaching and mentoring.** Teaches and mentors students, early-career researchers, and professionals through the FreeMoCap community channels, providing guidance on motion-capture methodology, data quality, and open research practice.